



DATA SCIENCE &
ARTIFICIAL INTELLIGENCE

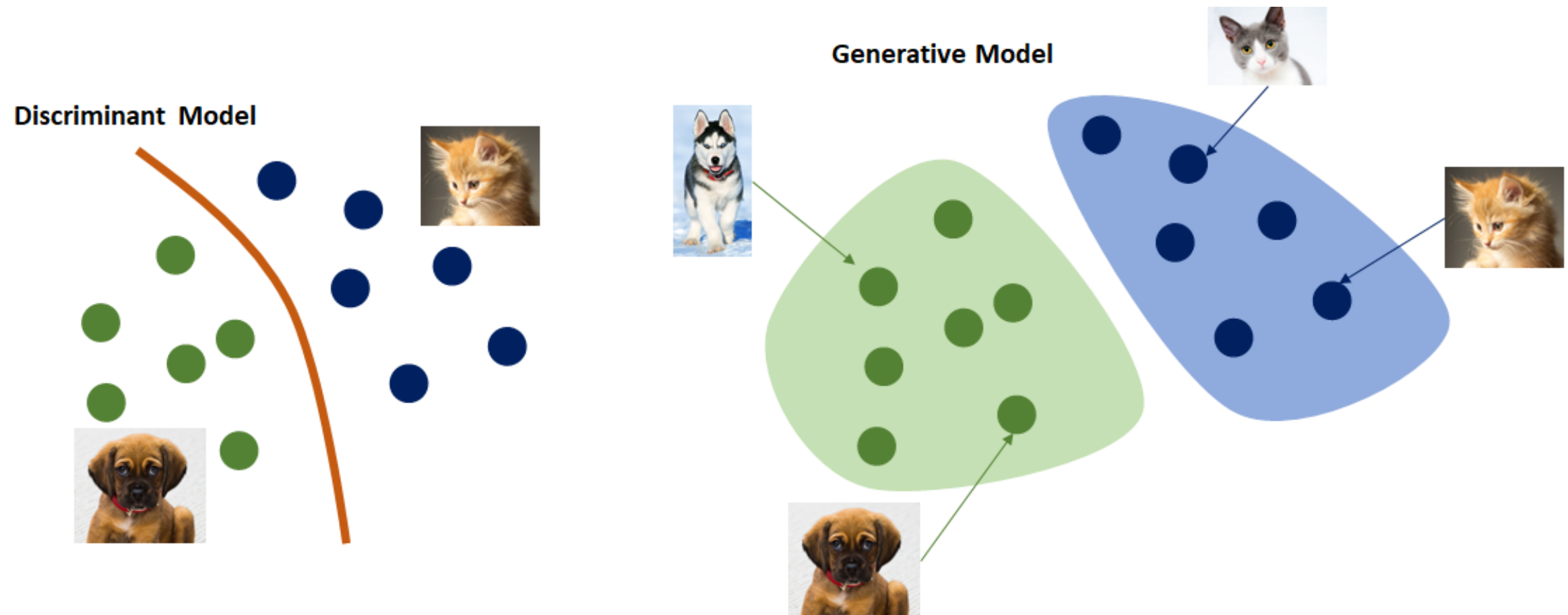
Probabilistic Machine Learning

Autoencoding Variational Bayes

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Generative Modelling

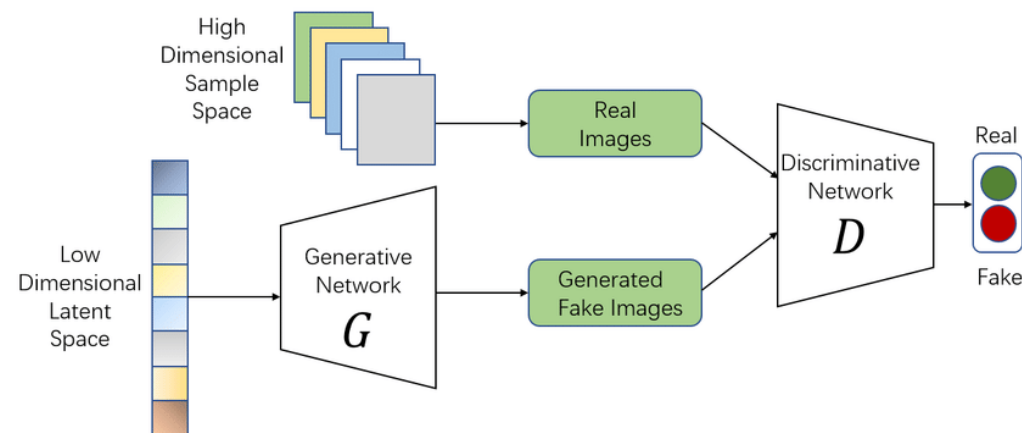
Goal: estimate the data distribution $p(x)$ by $p(x, \theta)$ - learning θ



Generative Modelling

Generative Adversarial Networks (GANs)

Variational Autoencoders (VAEs)

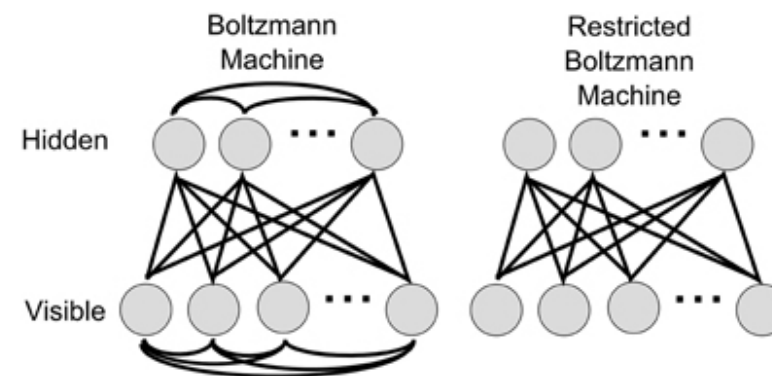


Distribution transformers

Normalizing flows

PGM-based

Boltzmann Machines



Noise-based

Energy-based Models

Diffusion Models

statistical physics based

Generative Modelling and Latent Variables

Introduce latent features z that describe high level properties of data

Assumptions:

$$p(x, z | \theta) = p(x | z, \theta)p(z)$$

1. $p(z | x)$ is intractable
2. working in the **big data** regime

Goals:

- a. Learning the parameters θ
- b. Approximating the posterior $p(z | x)$ in order to understand the latent features extracted from a given instance x
- c. Performing approximate inference on x to fill holes in an instance

Autoencoding Variational Bayes

Consider a parametric variational distribution $q(z|x, \phi)$

$$\begin{aligned}\mathcal{L}(\phi, \theta) &= \mathbb{E}_{q(z|x, \phi)}[\log p(x, z|\theta) - \log q(z|x, \phi)] \\&= \mathbb{E}_{q(z|x, \phi)}[\log p(x|z, \theta) + \log p(z) - \log q(z|x, \phi)] \\&= \mathbb{E}_{q(z|x, \phi)}[\log p(x|z, \theta)] - \mathbb{E}_{q(z|x, \phi)} \left[\log \frac{q(z|x, \phi)}{p(z)} \right] \\&= \mathbb{E}_{q(z|x, \phi)}[\log p(x|z, \theta)] - KL[q(z|x, \phi) || p(z)]\end{aligned}$$

Consider a mini batch $x_1, \dots, x_m$:

$$\mathcal{L}(\phi, \theta) \approx \frac{1}{m} \sum_{j=1}^m \mathbb{E}_{q(z|x_j, \phi)}[\log p(x_j|z, \theta)] - KL[q(z|x_j, \phi) || p(z)]$$

VAE ELBO

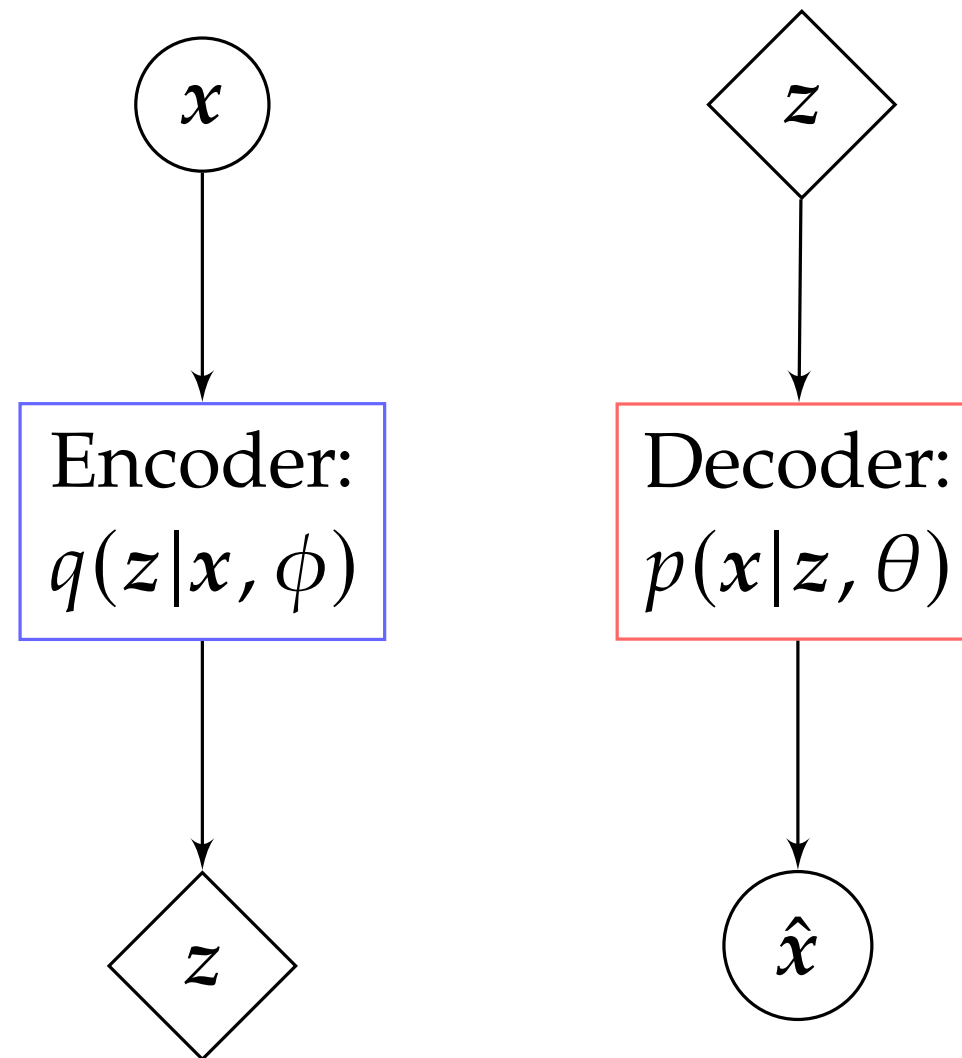
$$\mathcal{L}(\phi, \theta) \approx \frac{1}{m} \sum_{j=1}^m \mathbb{E}_{q(z|x_j, \phi)} [\log p(x_j|z, \theta)] - KL[q(z|x_j, \phi) || p(z)]$$

First term: reconstruction quality. Maximised when $p(x_j | z, \theta)$ assigns the highest probability to the original instance x_j

Second term: regularization. Forces the variational distribution to avoid being an identity map on the training data.

Amortised variational inference: parameters θ, ϕ are shared among all instances

VAE ELBO



Encoder: extracts latent features z from an input x

Decoder: reconstructs the input given the latent features

Reparametrization trick

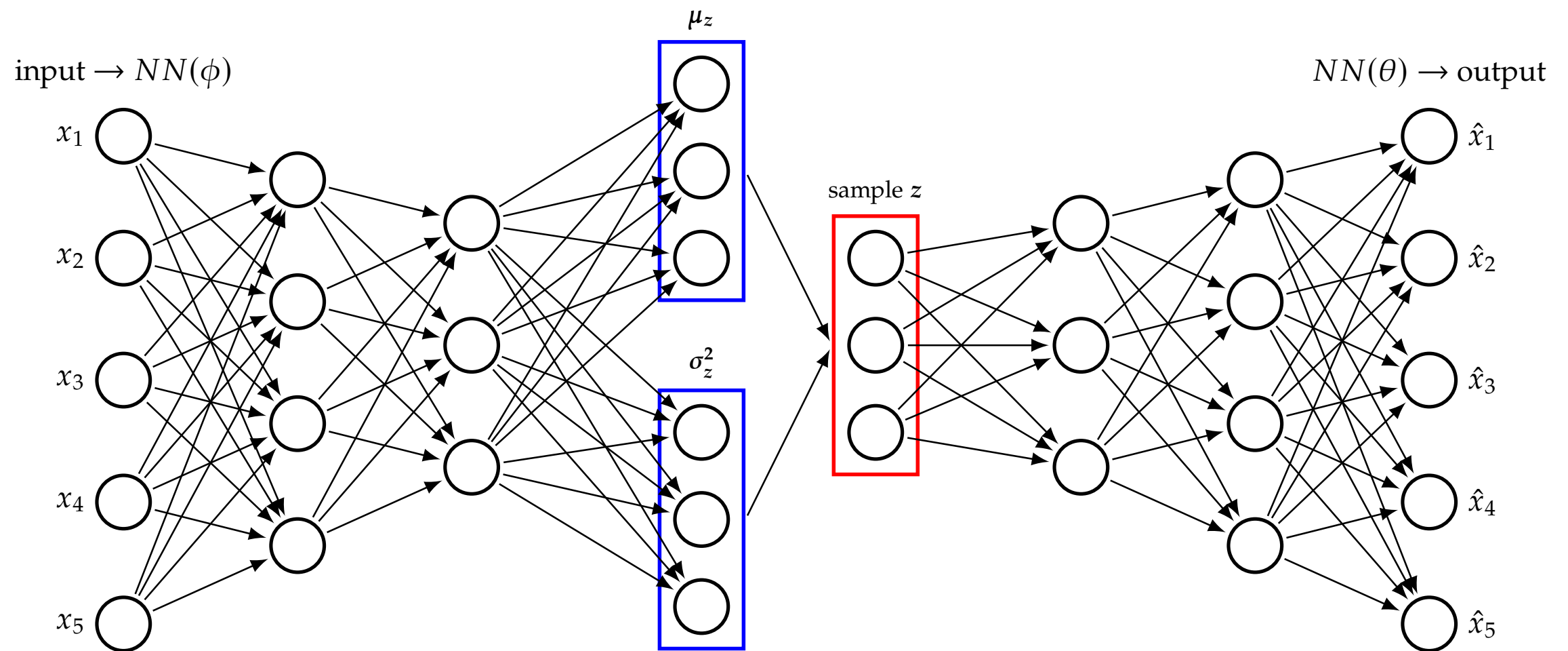
$$z = g(\varepsilon, x, \phi) \qquad \varepsilon \sim \mathcal{N}(0, I)$$

$$z = \mu_z(x, \phi) + \sigma_z^2(x, \phi) \cdot \varepsilon$$

$$\nabla_{\theta, \phi} \mathcal{L}(\phi, \theta) = \mathbb{E}_{\varepsilon} [\nabla_{\phi, \theta} \log p(x | g(\varepsilon, x, \phi), \theta)] - \nabla_{\phi} KL[q(z | x, \phi) || p(z)]$$

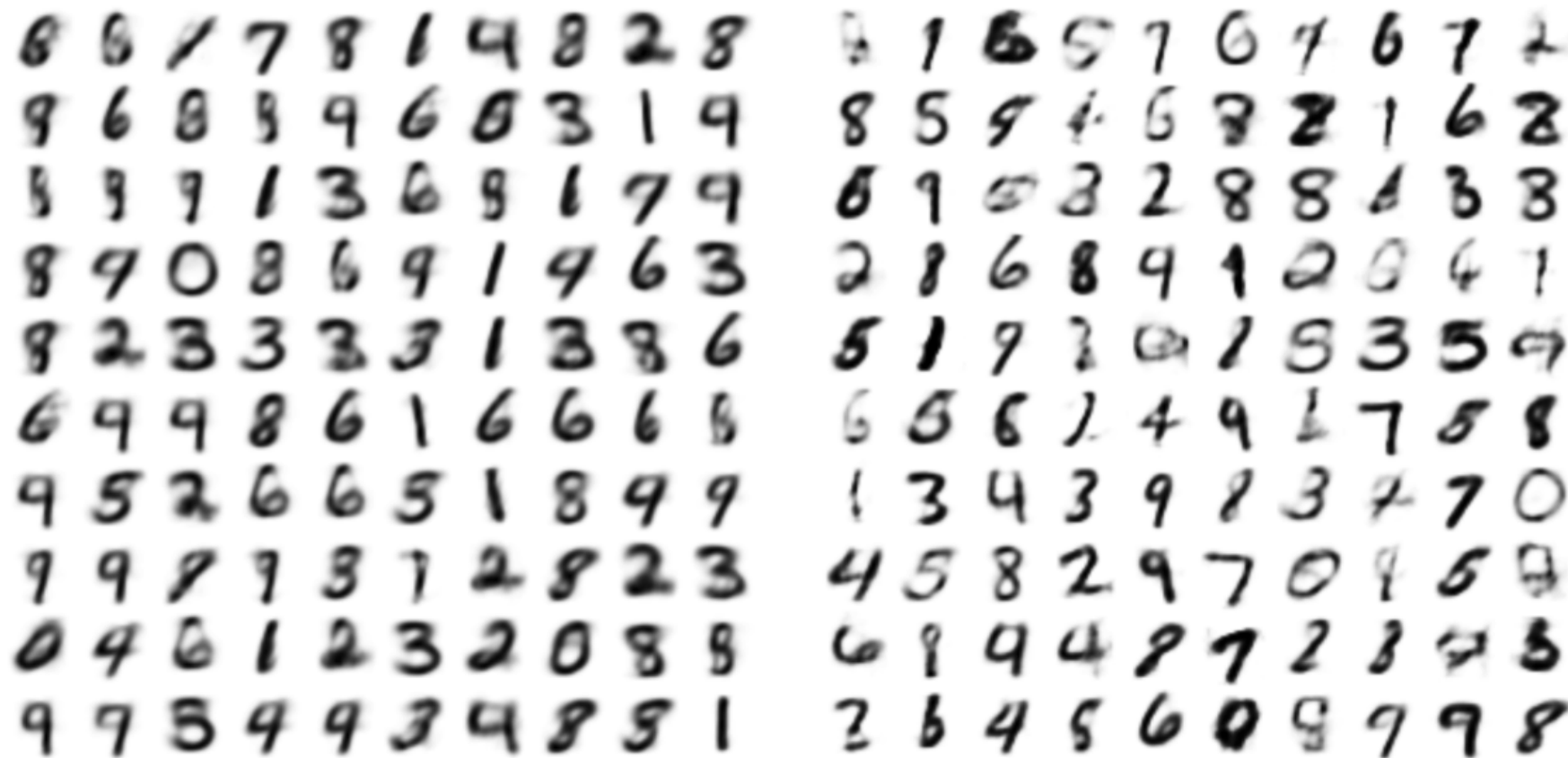
$\mu_z(x, \phi)$
 $\sigma_z^2(x, \phi)$ are represented as a neural network with params ϕ

Variational Autoencoder



We can learn it by stochastic variational inference!

Example: MNIST



(a) 2-D latent space

(b) 5-D latent space

Example: MNIST



(c) 10-D latent space



(d) 20-D latent space

